1st REVIEW

Blood Bank and Donor Locator App

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Under the Guidance
of

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GROUP NUMBER:20

Name of Candidate	Role
K ANUSRI	Maps, Notifications & Admin Module
J SIMRAN	Frontend / Android Developer
V RUKESH	Backend & Database Developer

ABSTRACT

- Android-based application connecting blood donors and recipients.
- Enables donors to register their blood group, location, and availability.
- Recipients can create blood requests and search for compatible nearby donors.
- Uses Google Maps SDK for location-based donor search.
- Uses MySQL database for storing donor, recipient, blood request, and donation history information.
- Uses Node.js and Express to provide REST APIs for communication between the Android application and database.
- Uses Firebase Cloud Messaging (FCM) for instant notifications.
- Maintains donation history and provides an admin panel.
- Main goal: reduce the time required to find suitable blood donors.

INTRDOUCTION

- Finding suitable blood donors during emergencies can be difficult.
- Traditional methods depend on phone calls, social media, or personal contacts.
- These methods may not provide real-time information about donor availability.
- The proposed app provides a centralized platform for donors and recipients.
- Users can search for compatible donors based on blood group and location.
- Notifications help donors respond quickly to urgent blood requests.

EXISTING SYSTEM

Blood donor information is often maintained through:

- Hospitals
- Blood bank
- Phone calls
- Social media groups
- Personal contacts
- Donor availability may not be updated regularly.
- Finding nearby compatible donors can take considerable time.
- Communication between recipients and donors is not centralized.
- Location-based donor searching is limited.

Existing systems:

1. AANJANA RAKT MITRA
2. UBLOOD

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PROBLEM

- Difficulty in finding compatible blood donors quickly.
- Lack of centralized and updated donor information.
- Difficulty identifying donors based on location and availability.
- Delayed communication between recipients and donors.
- Emergency situations require a faster and more reliable solution.

SOLUTION(Contribution)

- Develop an Android application connecting donors and recipients.
- Provide donor registration and verification.
- Allow recipients to create blood requests.
- Search for suitable donors using blood group and location.
- Use Google Maps to identify nearby donors.
- Send instant alerts through Firebase Cloud Messaging.
- Store donor information, requests, and donation history using MySQL through the backend API.
- Provide an admin panel for managing users and requests.

Objectives

- Connect blood donors with recipients through a centralized platform.
- Reduce the time required to find compatible donors.
- Enable location-based donor searching.
- Maintain updated donor availability.
- Provide instant notifications for blood requests.
- Maintain donation history.
- Improve communication between donors and recipients.
- Provide administrative control over donor and request information.

Modules

CO1 Database Design & SQL
Donor, Recipient, Blood Request, Blood Bank and Donation History tables; relationships, constraints, normalization and SQL queries using MySQL.

CO2 Backend APIs & Database Connectivity
REST APIs for donor registration, recipient management, blood requests, donor searching and database operations.

CO3 Backend APIs & Security
User authentication, blood request handling, donor-request management, API integration and database connectivity.

CO4 Node.js / Express
Backend server and REST APIs for managing users, donors, recipients and blood requests.

CO5 Microservices
Separate services for user management, donor/request management and notifications, if required.

CO6 Deployment & Documentation
GitHub, testing, documentation, architecture diagrams and final deployment.

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SOFTWARE REQUIREMENTS

Technology	Purpose
Android Studio	App Development
Java	Programming
XML	UI Design
MySQL	Backend Database
Firebase Cloud Messaging	Notifications
Google Maps	Location Services
Notion	Documentation
GitHub	Collaboration

CONCLUSION AND FUTURE SCOPE

Conclusion

- Provides a centralized platform for connecting blood donors and recipients.
- Enables faster identification of compatible and nearby donors.
- Improves communication through instant notifications.
- Helps manage donor information, requests, and donation history.

Future Scope

- AI-based donor matching.
- Hospital and blood-bank integration.
- Real-time donor availability.
- Emergency request prioritization.
- Expansion to more cities and regions.
- Integration with additional healthcare service.

THANK YOU